

# Lesson 3: Data Structures

A data structure is a way of organizing data so that it can be used efficiently. Unlike an array, a linked list does not store its elements in contiguous memory.

## 3.1 Stacks

A stack is a linear data structure that follows the Last In, First Out principle. For example, a pile of plates is a stack. Stacks are used to implement the undo feature in text editors.

Push adds an element to the top of the stack.

Pop removes the element at the top of the stack.

## 3.2 Queues

A queue is a linear data structure that follows the First In, First Out principle. For example, a line of customers at a bank is a queue. Queues are used to schedule jobs sent to a printer.

## 3.3 Algorithms and Complexity

An algorithm is a finite set of well-defined steps for solving a problem. The steps of problem solving are understanding the problem, planning a solution, coding the program and testing the result. Poor planning leads to programs that are hard to maintain.

Binary search is faster than linear search because it halves the search space at each step. The average of a list is computed as  $A = S / n$ , where S is the sum of the values and n is the number of values.

Ang algoritmo ay isang hanay ng mga hakbang para lutasin ang isang problema.